

Experiment Report

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Abstract

This study explores the fine-tuning of the LLaMA-3 model using the "unsloth" framework to optimize its performance for natural language processing tasks. The model was trained on the Stanford Alpaca dataset (different than original ruozhiba-llama3-tt)with a focus on instruction-following tasks. Key optimizations included the use of LoRA (Low-Rank Adaptation) techniques and gradient checkpointing to reduce memory usage while maintaining efficiency. Results demonstrate significant improvements in the model's ability to generate contextually accurate responses, validating the effectiveness of the proposed fine-tuning method.

Introduction

Large-scale language models like LLaMA have transformed natural language processing (NLP) tasks, showing remarkable performance across various domains. However, adapting these models for specific tasks remains an essential step to further enhance their capabilities. This work investigates the fine-tuning of the LLaMA-3 model, leveraging the lightweight "unsloth" framework. By applying LoRA-based optimization techniques and training the model on the Stanford Alpaca dataset, we aim to improve the model's instruction-following abilities, offering new insights into the fine-tuning of large models for specific task-oriented applications.

Materials and Methods

Model: LLaMA-3 with 8 billion parameters, quantized to 4-bit for reduced memory consumption.

Dataset: I used two datasets: **kigner/ruozhiba-llama3-tt** and **hibing624/alpaca-zh**, and tested the differences in their responses. **kigner/ruozhiba-llama3-tt** is primarily optimized for Chinese dialogue systems or language generation. It likely covers more complex text comprehension and adjustments for specific conversational styles in certain domains. This dataset may be more suited for interactions involving cultural nuances or domain-specific knowledge, such as in professional fields or technical jargon. **hibing624/alpaca-zh**, on the other hand, is based on the Chinese version of the Alpaca dataset. It references self-instruct data derived from GPT-4 following the Alpaca method. The dataset is mainly used to fine-tune models for instruction-following tasks, covering a wider range of general tasks like answering questions, summarizing information, or executing specific commands. It is designed to improve the model's versatility and task flexibility in Chinese environments.

Framework: The "unsloth" framework was employed for model fine-tuning, which supports efficient model adaptation and training.

Training Configuration:

- LoRA (Low-Rank Adaptation) was applied to specific modules (q_proj, k_proj, etc.) with a rank of 16 and dropout set to zero.
- Gradient checkpointing was enabled to optimize memory usage, particularly for handling long sequences.
- Key parameters included a batch size of 2, a learning rate of 2e-4, and a maximum of 60 training steps.
- Training was conducted using the AdamW optimizer with an 8-bit precision setting to further reduce memory footprint.

Results

Below are the test results from experimenting with the pre-trained unsloth/llama-3-8b-bnb-4bit model.

Instruction	Input	Output
只用中文回答問題	如果有人偷了麵包，你會怎麼做？	我會告訴他，我沒有麵包
只用中文回答問題	如果有人偷了麵包，你會怎麼做？	如果有人偷了麵包，你會怎麼做？

It can be observed that the model primarily repeats my input or fails to address the actual question. Therefore, we proceeded with training using two datasets. Below are the test results, with the upon side representing the **kigner/ruozhiba-llama3-tt** dataset and the below side representing the results from the **hibing624/alpaca-zh** dataset after training.

Instruction	Input	Output
只用中文回答問題	如果有人偷了麵包，你會怎麼做？	我會去報警或者找麵包店店員，因為這是一個犯罪行為。
只用中文回答問題	如果有人偷了麵包，你會怎麼做？	如果有人偷了我的麵包，我會立即報警，並且提供相關訊息給警察。

Below are the results from testing the model's ability to handle unfamiliar concepts.

Instruction	Input	Output
只用中文回答問題	什麼是吉伊卡哇？	吉伊卡哇是一種來自於印度的美食
只用中文回答問題	什麼是吉伊卡哇？	吉伊卡哇（英語：Jai Kaliwa）是印尼的傳統遊戲，類似於西方的棒球和排球。吉伊卡哇遊戲的規則比較簡單，主要是利用球拍來擊打一球，擊打到對方的球網內就可以得分。吉伊卡哇遊戲通常在比賽中由兩隊人數相同的球員進行對抗，比賽通常持續數分鐘到數小時不等。

Conclusion

In this experiment, it was observed that when encountering unfamiliar terminology, the LLM tends to fabricate responses. However, by fine-tuning the LLaMA model with a custom-defined training dataset, it is possible to adapt the model without the need to train a large LLM from scratch. This allows for customization of the model to better suit specific needs. In the future, given the opportunity, I hope to design a custom LLM tailored specifically for answering specialized questions in the domain of 吉伊卡哇.

References

1. AI Discovery 2045. (n.d.). AI Discovery 2045 YouTube Channel. Retrieved from <https://youtube.com/@aidiscovery2045>